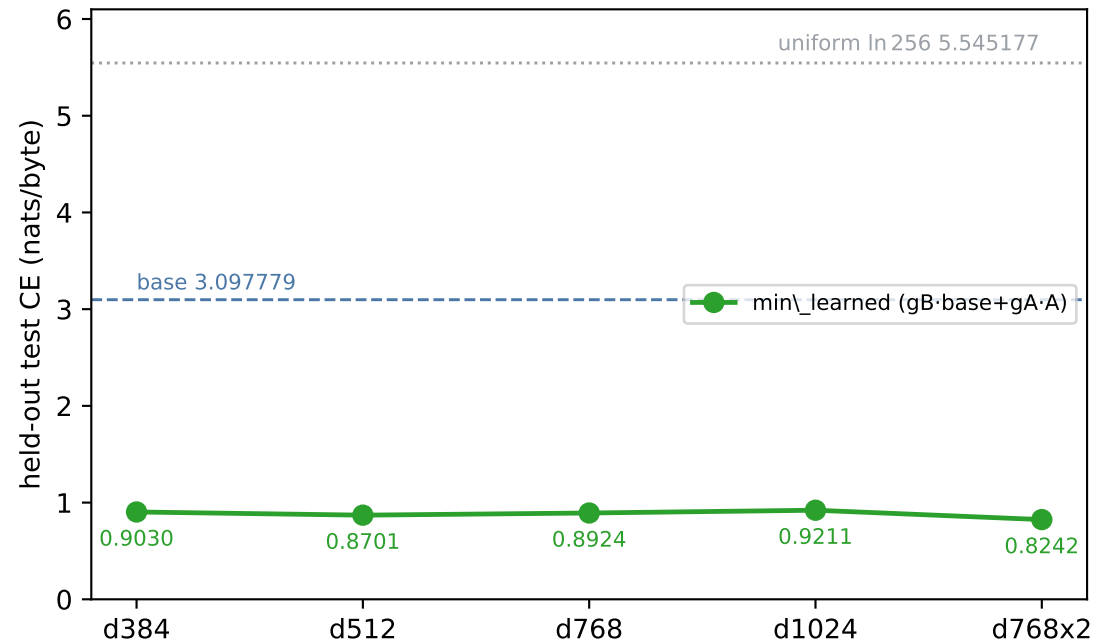


OMEGA minimal-gate 5-rung scale ladder (leak-free, causal_ca=True; self-test 0.000 every rung)

(a) minimal-gate HOLDS at every rung
 $\text{min_learned} \ll \text{base} < \text{uniform}$



(b) advantage is SCALE-STABLE
48M $\rightarrow$ 335M params, Δ flat

